

Stone Duality

Connecting Algebra and Topology via Logic

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Link to course website:

levinhornischer.github.io/StoneDuality_ESSLLI/

WELCOME!

- ▶ This course is an introduction to **Stone duality**.
- ▶ This is an exciting area of **logic** at the intersection of math, computer science, and philosophy.
- ▶ **Stone's theorem** says that certain algebras (Boolean algebras) are in a precise sense equivalent to certain topological spaces (now called Stone spaces).
- ▶ Behind this mathematical fact lies a deep conceptual insight: that the algebraic and the spatial perspective are just **two sides of the same coin**!
- ▶ Understanding this is the goal of the course.

- ▶ **Schedule:** in the first week of ESSLLI 2026 (August 3–7, 2026), from 09:00 to 10:30 in room C206.
- ▶ **About me:** I'm Levin Hornischer, an assistant professor at the Munich Center for Mathematical Philosophy (MCMP) at LMU Munich. Before that, I was at the Institute for Logic, Language and Computation (ILLC) at the University of Amsterdam, where I also did my PhD.
- ▶ **Comments and feedback** are most welcome: just email me at Levin.Hornischer@lmu.de.
- ▶ Please interrupt for **questions any time!**

- ▶ There is a website for the course:

levinhornischer.github.io/StoneDuality_ESSLI/



- ▶ These slides are intended as **handouts** for the lecture (hence more cluttered than presentation slides—sorry!).
- ▶ I have given a full semester course on Stone duality, on which this course is based. You find the **lecture notes** linked on the course website.
- ▶ **Prerequisites:** An introductory course in logic (specifically classical propositional logic) and some familiarity with mathematics (ideally, but not necessarily, having seen elementary concepts of topology and algebra).

- ▶ A wonderful recent textbook on Stone duality is [GG24] and further great textbooks on—or closely related to—Stone duality include [BD75; DP02; Vic89; GH08; Giv14; Grä11; Grä03; AJ94].
- ▶ Research monographs on duality theory and related topics include [Joh82; Gie+03; DST19; Gou13; PP12; ORR15].
- ▶ Extensions of Stone duality from propositional to first-order logic include [Mak87; Lur20; AF13].
- ▶ Stone duality also is an excellent pathway into category theory [Lei14; Smi25], and in particular categorical logic [Pit01] and topos theory [Vic07].

PLAN

1. Introduction and motivation: the idea of a duality
2. Boolean algebras
3. Stone spaces
4. Stone's theorem and proof idea
5. Applications and generalizations

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THE GENERAL IDEA OF A DUALITY

- ▶ The general idea of a duality is that there are two, in a sense complementary perspectives on the same underlying phenomenon.
- ▶ Metaphorically speaking, there are **two sides** (the perspectives) of the **same coin** (the phenomenon).
- ▶ This occurs throughout the sciences. We consider four brief examples. (Also see [Geh09; Abr91; Vic89].)

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- ▶ Now we all know it, but it is remarkable: there is a close correspondence between algebra and geometry.
- ▶ Algebraic objects, like the polynomial $x^2 + y^2 = 1$, correspond to spatial objects, in this case to the unit circle.
- ▶ We can solve geometric problems, e.g., which ratios and angles can be constructed with a compass and straightedge, by translating them to the algebraic side and then using algebraic methods (here Galois theory).
- ▶ The duality of algebra and geometry is at the heart of algebraic geometry and bears quite some similarities to Stone duality [Joh82; MM92; Vic07].

- ▶ On the one hand, a physical system is described by its **state space**, i.e., the collection of states that the system could be in.
- ▶ On the other hand, the system is also described by the **observations** we can make about it.
- ▶ We would hope that these two perspectives on the system are ‘dual’ to each other:
 - ▶ The states should be **determined** by the observations that they give rise to; and
 - ▶ the observations should be **determined** by the states that give rise to them [e.g. **Str08**, p. 24].
- ▶ The formal duality (e.g., Gelfand–Naimark) is different in detail from Stone duality but similar in spirit. (For more, see, e.g., [**Rue11**; **RBW15**; **WW23**; **DB25**].)

- ▶ Computer programs are written in a programming language. Much like for sentences in a natural language, we can ask what their meaning is.
- ▶ The meaning of a program is called its **denotation** and is, roughly, the input-output function that it computes.
- ▶ But we can also describe the program by its **program logic**:
 - ▶ e.g., given an even number as input, the program outputs a prime number.
 - ▶ More generally: **Hoare triples** ‘if currently P holds, then running the program C will result in Q being satisfied’.
- ▶ The far-reaching insight of Abramsky [Abr91] was that we again should have a duality between the denotational semantics and the program logic.

- ▶ The central question of philosophy of language is: what is the **meaning** of sentences?
- ▶ One prominent view is **possible worlds semantics**: The meaning of a sentence is the set of **possible worlds** in which the sentence is true.
- ▶ On a broadly **inferentialist** position, we describe sentence meanings by how they relate to each other: which implies which, what conjunctions are, etc. This data is recorded in the set of all sentence meanings together with the implication relation, conjunction, negation, etc.—aka the **algebra of propositions**.
- ▶ Duality talk is surprisingly scarce in philosophy, but given the success in other fields it holds much promise!

<i>Field</i>	<i>Phenomenon</i>	<i>Space</i>	<i>Algebra</i>
Mathematics	Geometry	Spatial object	Polynomial
Physics	Physical system	State space	Observations
Computer Science	Program	Denotation	Program logic
Philosophy	Meaning	Possible worlds	Propositions

- ▶ **Algebra:** Set of observations, propositions, etc. with **operations** (e.g., conjunction $\wedge$) that satisfy certain **laws** (e.g., $a \wedge b = b \wedge a$).
 - ▶ A specific type of such an algebra we will make precise as a Boolean algebra.
- ▶ **Space:** Set of **points** (e.g., states, possible worlds, etc.) with some **spatial structure**, e.g., a notion of distance between points.
 - ▶ A specific type of such a space we will make precise as a Stone space, which is a particular kind of topological space.

KEY IDEA OF DUALITY

- ▶ **Duality**: A bijective correspondence between the class of spaces and the class of algebras under consideration, i.e.,
 - ▶ a method to translate a space into an algebra, and
 - ▶ a method to translate an algebra into a space,
 - ▶ such that translating forth and back is isomorphic to where one started.
- ▶ In this sense of **intertranslatability**, the spatial perspective is equivalent to the algebraic perspective and **both describe the same phenomenon**.
- ▶ In our specific case, **Stone's theorem** makes this precise as: the category of Boolean algebras is dually equivalent to the category of Stone spaces.
- ▶ Such a duality not only provides a deep **conceptual link** between the two perspective. It also can help practically by **translating a problem** from one side to the other, where different methods and intuitions are available (cf. again geometry and algebra).

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EXAMPLE: POSSIBLE WORLDS VS PROPOSITIONS I

We do not have the precise concepts yet, but let's sketch how the duality looks like in the philosophy example.

- ▶ Translation from spaces to algebras:
 - ▶ Assume a space X of possible worlds. According to possible worlds semantics, a proposition a is a set of possible worlds, i.e., $a \subseteq X$.
 - ▶ Conversely, is every subset $a \subseteq X$ a proposition? This is not commonly discussed, but we will see: only if a is “closed under [similarity](#)” (technically: a [clopen](#) subset in the topology on X).
 - ▶ Proposition a implies proposition b iff in all worlds $x \in X$ in which a is true, also b is true, i.e., $a \subseteq b$.
 - ▶ The conjunction of two propositions a and b simply is the intersection $a \cap b$. Similarly for other operations.
 - ▶ So the space X translates to the [algebra](#) $\text{Clp}(X) = (A, \subseteq, \cap, \dots)$, where A is the set of all $a \subseteq X$ that are “closed under similarity”.

EXAMPLE: POSSIBLE WORLDS VS PROPOSITIONS II

- ▶ Translation from algebras to spaces:
 - ▶ Assume an algebra $A = (A, \leq, \wedge, \dots)$ of propositions.
 - ▶ According to [Ersatzism](#), given a language/propositions, we can construct ‘substitute possible worlds’ as maximally consistent sets of propositions.
 - ▶ So an Ersatz possible world is a set $F \subseteq A$ such that
 - (1) F is closed under implication and conjunction and does not contain the logical contradiction $\perp$, and
 - (2) for any other such set $G \subseteq A$, if $F \subseteq G$, then $F = G$.(We will call such an F a [maximal filter](#).)
 - ▶ Let $\text{Spec}(A)$ be the space of all Ersatz possible worlds (we will see that there is a natural way to define a topology on $\text{Spec}(A)$).

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EXAMPLE: POSSIBLE WORLDS VS PROPOSITIONS III

- ▶ Translating a space X of possible worlds to the algebra $\text{Clp}(X)$ and then back to Ersatz space $\text{Spec}(\text{Clp}(X))$ yields a space “isomorphic” to X .
 - ▶ We need, in particular, a bijection $f : X \rightarrow \text{Spec}(\text{Clp}(X))$.
 - ▶ We map a world x to the set $f(x) = \{a \in \text{Clp}(X) : x \in a\}$ of all (possible worlds semantics) propositions that x makes true.
- ▶ Translating an algebra A to Ersatz worlds $\text{Spec}(A)$ and then back to the algebra $\text{Clp}(\text{Spec}(A))$ yields an algebra isomorphic to A .
 - ▶ We need, in particular, a bijection $h : A \rightarrow \text{Clp}(\text{Spec}(A))$.
 - ▶ We map a proposition a to the set $h(a) = \{F \in \text{Spec}(A) : a \in F\}$ of Ersatz worlds that make true a .

Now we develop these ideas formally.

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